

SOWBHAGYA MOHANTHY

AI Engineer & Software Developer · IBM · Hyderabad, India

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AI Engineer · AI Pipeline Lead · Full-Stack Developer

Results-driven Software Developer & AI Pipeline Lead with 3 years at IBM. Conceived and led production AI systems end-to-end — from semantic clustering pipelines and LLM orchestration to OpenShift deployment. Hands-on expertise building Autonomous AI Agents with OpenAI Agents SDK, CrewAI, LangGraph, AutoGen, and MCP. Skilled across the full stack: Python, FastAPI, React, Node.js, MongoDB, PostgreSQL.

CORE SKILLS

Agentic AI

OpenAI Agents SDK · CrewAI · LangGraph · AutoGen · MCP · Multi-Agent Systems · Human-in-the-Loop · Agent Orchestration · LangSmith · Gradio

AI / ML

LLMs · RAG · IBM watsonx.ai · UMAP · HDBSCAN · c-TF-IDF · Random Forest · PCA · Semantic Embeddings · Ensemble Classification · Incremental ML · Cosine Similarity

LLM Engineering

IBM Granite · Gemini · LangChain · Prompt Engineering · Structured Outputs · Token Budget Management · Context Window · Multi-batch Fan-in Synthesis

Backend / APIs

Python 3.12 · FastAPI · APScheduler · Pydantic v2 · Node.js · Express.js · REST APIs · Microservices · Uvicorn · Flask · Django

Frontend

React.js · Next.js · Redux · JavaScript · Playwright · Gradio · Streamlit

Databases

MongoDB · PostgreSQL · Milvus DB · Redis · FAISS · ChromaDB

DevOps / Cloud

IBM Cirrus · OpenShift · Docker · CI/CD · Git · Jenkins · Poetry · Pytest · Ruff · mypy

WORK EXPERIENCE

Software Developer — AI & Full-Stack

IBM · CIO Agile & Emerging Technologies · Bangalore, India (Hybrid) · July 2023 – Present

- Conceived, architected, and led end-to-end development of Recommendation Engine — a production AI microservice converting raw enterprise customer feedback (reviews, NPS surveys, sales conversations) into team-routed business recommendations; deployed on IBM Cirrus/OpenShift, delivering weekly insights to Product, Sales, Marketing & Pricing teams.
- Designed a 5-stage automated AI pipeline: LLM batch summarization + IBM Granite embeddings → UMAP+HDBSCAN semantic clustering (384D→12D, c-TF-IDF auto-labeling) → 3-tier ensemble classifier (Rule-Based ×0.4 + Random Forest ×0.6 + IBM Granite LLM fallback) → 128K-context token-budget summarization → cosine-deduplicated recommendations (0.92 threshold) across 4 date windows (4w / quarter / 6m / 12m).
- Built incremental clustering via `hdbscan.approximate_predict` on versioned .pkl models — new feedback assigned without retraining; reduced weekly compute cost significantly.
- Engineered 3-tier ensemble classifier with Jensen-Shannon divergence disagreement detection; V2 scores all 4 teams simultaneously with High/Medium/Low confidence tiers — 4 MongoDB documents per product per pipeline run.
- Built production FastAPI microservice with APScheduler cron pipeline, per-step toggle flags, MongoDB connection pooling + bulk-upsert across 10+ collections; CI/CD via cirrus-deployment, Slack build notifications.
- Developed AI-powered chatbots with LLMs, RAG, and Agentic AI (WatsonX) — ~40% improvement in automated query resolution across enterprise systems.
- Built self-healing Selenium test framework using LLMs + neural networks to auto-update scripts when UI elements change — cut manual QA effort by 70%.
- Engineered high-performance retrieval system with Milvus DB + Redis — ~25% faster response times.

- Delivered scalable full-stack apps: React, Node.js, Express, Python, PostgreSQL/MongoDB with end-to-end CI/CD.

Software Developer Intern

IBM · CIO Agile & Emerging Technologies · Bangalore, India (Hybrid) · Jan 2023 – July 2023

- Developed and tested REST APIs using Node.js, Express, and MongoDB.
- Built dynamic UI components with React.js and Redux, improving user experience and performance.
- Wrote unit tests to enhance code reliability and reduce bug frequency by ~20%.
- Managed Git-based version control across teams enabling seamless collaboration.

PROJECTS

Virtual Twin — Personal Agentic AI

LangGraph · Gemini · ChromaDB · Streamlit · SendGrid · Google Calendar API · RAG

- Built an agentic AI virtual twin of myself — answers questions about skills, experience, and projects using RAG over resume and a 570-line deep profile document.
- Designed a LangGraph supervisor + 4 specialised ReAct sub-agents: RAG Agent (ChromaDB retrieval), Scheduler Agent (multi-turn Google Calendar meeting booking), Notifier Agent (SendGrid email alerts for knowledge gaps), and Chitchat Agent.
- Supervisor uses keyword-first routing (zero LLM tokens) with Gemini fallback; session state persisted via InMemorySaver — zero-infra, runs on Streamlit Community Cloud.
- Knowledge base: 161 chunks from 3 sources (resume PDF + deep profile + inline FAQ), embedded with gemini-embedding-001 and stored in ChromaDB.

Recommendation Engine AI | Production · IBM CIO Engineering

Python 3.12 · FastAPI · APScheduler · IBM watsonx.ai · IBM Granite · UMAP · HDBSCAN · scikit-learn · MongoDB

- Flagship production system: conceived architecture, led team, shipped to production — weekly AI-generated insights for 4 business teams.
- UMAP (384D→12D cosine) + HDBSCAN EOM + c-TF-IDF bigram labeling for unsupervised semantic clustering of multi-source enterprise feedback.
- 3-tier ensemble classifier + cosine deduplication (0.92) + top-5 flagging across 4 date windows.

AI-Driven Selenium Self-Healing Test Framework

Python · LLMs · Neural Networks · Selenium · WebDriver

- LLM + neural network system that auto-updates Selenium scripts when UI elements change or are deleted — 70% reduction in manual QA effort.

RAG-Based Intelligent Assistant

LangChain · Milvus DB · WatsonX · Redis

- Multi-step retrieval, semantic embeddings, and WatsonX model integration for high-accuracy enterprise knowledge querying.

EDUCATIONAL BACKGROUND

Bachelor of Technology | CSE | GPA 9.2 / 10

Gokaraju Rangaraju Institute of Engineering & Technology · Aug 2019 – June 2023

Higher Secondary Education (Science)

Narayana Junior College · 2017 – 2019

High School Diploma

Navodaya High School · 2016 – 2017

CERTIFICATIONS

- IBM WatsonX Essentials — IBM · 2024
- IBM Agile Explorer — IBM
- Enterprise Design Thinking Practitioner — IBM
- IBM Developer Jumpstart — Practitioner — IBM
- CCNA: Introduction to Networks — Cisco · 2022
- AWS Academy Cloud Foundations — Amazon Web Services · 2021